
Invocation-Level Reliability of Tool-Using Agents

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Abstract

1 An early tool-call error can corrupt every downstream input even when an agent
2 subsequently applies the right local rule. We isolate this effect with matched
3 evaluations under a correct baseline history and the model’s own free-running
4 history. For tasks of depth d , their pooled invocation rates $\bar{p}_d$ and $\bar{g}_d$ define a fit-
5 free relative propagation loss $L_d = 1 - \bar{g}_d/\bar{p}_d$. Across five open-weight models on
6 controlled routing tasks, the two 7–8B models lose about 68% of baseline capability
7 by depth 6; stronger models mostly remain at ceiling. Our main result is a construct-
8 validity condition for trajectory scoring. Let Y_k^* be the canonical next target after a
9 divergence and H_k the agent-observable history. If Y_k^* has conditional guessing
10 probability at most ϵ_k given H_k , then any fixed-gold scorer credits the agent with
11 probability at most ϵ_k , regardless of its local competence. Gold-agreement severity
12 is therefore driven to a scorer-imposed boundary, and observed reconvergence
13 cannot measure behavioral recovery. **Fixed reference alone is not sufficient:** the
14 result applies when the canonical target becomes unreachable or unpredictable,
15 and excludes reconstructible targets and evaluators that accept alternative states
16 or milestone paths. Our hidden-constant tasks instantiate the condition: 0 of
17 869 corrupted-context calls match gold, and 0 of 580 opportunities reconverge.
18 Conditional-on-state replay instead asks whether a call correctly continues from
19 the state the agent actually holds; it yields interior severity estimates of 0.149
20 and 0.316 and requires no new model queries. Alternative absolute-gap and error-
21 amplification summaries give the same qualitative depth trend.

1 Introduction

23 Language model agents interact with external systems by emitting structured *tool calls*: each call
24 names a function (e.g. `get_weather`, `query_database`) and fills its required arguments. A call
25 can fail by selecting the wrong function (*selection error*) or by providing incorrect argument values
26 (*argument error*). In a multi-step task where each call’s output feeds into the next, a single wrong call
27 early in the chain corrupts the inputs to every downstream call, even if the model reasons correctly
28 from that point on.

29 Many benchmarks report task success or agreement with a canonical call, without separately mea-
30 suring selection, arguments, and the downstream effect of an earlier error. State-based benchmarks
31 allow alternative solutions, and process benchmarks annotate individual steps, but neither directly
32 supplies the matched baseline/free-running contrast studied here. These distinctions matter because
33 local incompetence, state corruption, and ordinary later-step difficulty require different interventions.

34 We introduce a protocol that makes this distinction. A call is *correct* when it selects the appropriate
35 tool and supplies the arguments required by the scoring regime. We measure it under two matched
36 conditions: a *baseline* condition with a guaranteed-correct history and a *free-running* condition
37 with the model’s own prior outputs. Their gap measures the loss associated with accumulated state
38 corruption while controlling for the prompts and later-step difficulty present in the baseline arm.

The study began as an empirical comparison and exposed a construct-validity problem. When divergence makes the canonical next target unpredictable from everything the agent observes, gold agreement measures target reachability rather than the competence of the next call. This is not a property of every fixed-reference benchmark: it depends on an explicit unreachable-target condition. We state that condition, give a counterexample, and separate the resulting scorer-imposed boundary from the model’s behavior.

Contributions.

1. A disaggregated measurement protocol that separately scores tool selection and argument correctness under matched baseline and free-running conditions, with a routing-task design where selection errors, not just argument errors, propagate.
2. A formal unreachable-target condition under which canonical gold agreement is bounded by the target’s conditional guessing probability, making gold-based severity and reconvergence scorer-dominated rather than behavioral.
3. A remedy, conditional-on-state scoring, where each call is judged against what would be correct given the value the model actually holds: implemented and applied retrospectively to cached completions at zero marginal cost.
4. A controlled error-injection check that isolates the causal effect of state corruption from the confounds present in the observational baseline/free-running contrast, finding the observational conditional-severity estimate is inflated by roughly $2.9\times$ and that severity is confined to the selection channel.
5. Robustness summaries beyond L_d (absolute excess error and error amplification), plus per-model diagnostics for rule discrimination and error position.

2 Related Work

Toolformer [1], ReAct [2], and Gorilla [3] established tool-augmented agents. ToolLLM [4], StableToolBench [5], API-Bank [6], and BFCL [7] emphasize task success or agreement with reference calls. RoTBench [9] and MTU-Bench [10] separate selection from argument accuracy; FuncBenchGen [11] generates fresh dependency-graph tasks and reports stale-value failures. Our distinct estimand is the matched change in invocation correctness when gold history is replaced by the same model’s history.

Not all agent evaluation uses one canonical trajectory. τ -bench [8] grades final environment state; AppWorld [16] uses state-based unit tests that permit multiple solutions; ToolSandbox [17] evaluates intermediate and final milestone DAGs over arbitrary trajectories. These methods avoid the unreachable-canonical-target failure but primarily measure outcome or milestone completion, not local post-divergence competence. Section 5.3 states the narrower condition under which canonical-call scoring becomes scorer-dominated.

Process-oriented work is complementary. AgentProcessBench [18] provides human step labels and an explicit propagation rule, while FALAT [19] traces dependencies to separate decisive errors from inherited downstream failures. Relign [12], Hammer [13], and representation-level detection [14] instead reduce or detect the production of an incorrect call. We intervene on history to quantify propagation and compare gold-agreement with conditional-on-state scoring.

3 Method

3.1 Definitions

A task of dependency depth d contains calls indexed by position $k \in \{1, \dots, d\}$. For task i , let $B_{i,k}$ and $F_{i,k}$ indicate invocation correctness in the baseline and free-running arms. With N_d tasks at depth d , the reported depth-level estimands are

$$\bar{p}_d = \frac{1}{N_d d} \sum_{i=1}^{N_d} \sum_{k=1}^d B_{i,k}, \quad \bar{g}_d = \frac{1}{N_d d} \sum_{i=1}^{N_d} \sum_{k=1}^d F_{i,k}. \quad (1)$$

- **Pooled baseline correctness** $\bar{p}_d$: invocation accuracy with a guaranteed-correct prior history.
- **Pooled free-running correctness** $\bar{g}_d$: invocation accuracy with the model’s own prior outputs.

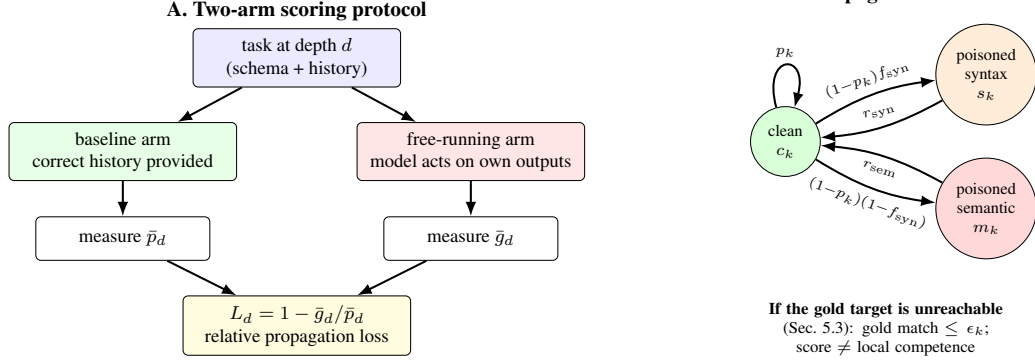


Figure 1: (A) Matched evaluation at task depth d : pooled baseline and free-running invocation rates define L_d . (B) A separate step-indexed propagation model. Under the unreachable-target condition of Section 5.3, canonical gold agreement is bounded by target predictability rather than local competence.

- **Relative propagation loss** $L_d := 1 - \bar{g}_d / \bar{p}_d$. We also report the absolute gap $A_d = \bar{p}_d - \bar{g}_d$ and error amplification $E_d = (1 - \bar{g}_d) / (1 - \bar{p}_d)$.

L_d is descriptive and invocation-pooled; it is not the stepwise quantity in the Markov recurrence below. Calls from one task are dependent, so all reported intervals use paired task-level bootstrap resampling: each of 2,000 draws samples N_d tasks with replacement, includes all positions from each selected task in both arms, recomputes $\bar{p}_d$, $\bar{g}_d$, and L_d within the replicate, and pairs baseline/free-running observations by task identity. The 89% level (5.5th and 94.5th percentiles) matches PyMC’s default highest-density convention used elsewhere in this work. When a replicate draws only tasks with $\bar{p}_d = 0$, L_d is undefined and that replicate is excluded; this occurred in fewer than 0.1% of replicates at any depth.

3.2 A propagation model with recovery

For interpretation only, we model step-indexed context as clean, poisoned-by-syntax, or poisoned-by-semantics (Figure 1B). Let p_k be correctness at position k under clean history and $f_{syn}(k)$ the syntactic share of new errors. With origin-specific recovery rates r_{syn} , r_{sem} and poisoned-state correctness $\rho_k = (1 - \pi)p_k$, occupancies evolve as

$$c_{k+1} = c_k p_k + s_k r_{syn} + m_k r_{sem}, \quad s_{k+1} = c_k (1 - p_k) f_{syn}(k) + s_k (1 - r_{syn}), \quad (2)$$

$$m_{k+1} = c_k (1 - p_k) (1 - f_{syn}(k)) + m_k (1 - r_{sem}). \quad (3)$$

The model-implied step accuracy is $g_k^{\text{model}} = c_k p_k + (s_k + m_k) \rho_k = p_k (1 - \pi x_k)$ for poisoned mass $x_k = s_k + m_k$. We fit π, r_{syn}, r_{sem} hierarchically (PyMC, NUTS) as a cautionary diagnostic, not as the estimator of L_d . Section 5.3 explains why gold-agreement data do not identify the intended behavioral interpretation of these parameters under an unreachable target.

3.3 Task design

Each task is a hidden dependency graph over deterministic executable functions, giving exact ground truth per call and direct control of depth ($\{1, 2, 4, 6, 8\}$). Tool outputs lie in a size- M space ($M = 100,000$) and take the form $(a \cdot \arg + b) \bmod M$, with per-tool constants omitted from the schema. Section 5.3 does not infer unpredictability from secrecy alone; it states the conditional target-predictability assumption required for the bound.

Routing tasks (primary): the correct next tool is a parity rule applied to the incoming value, not a pre-announced order, so a wrong choice changes what is correct downstream. Selection is scored *conditionally*, against the tool correct given the value the model actually holds, with divergence from gold recorded separately, so applying the rule correctly to a corrupted input is not misclassified as selection failure. **Linear tasks (control):** order announced, argument copied verbatim. Pilot

runs were degenerate ($\bar{p}_d = \bar{g}_d = 1.000$ on two models), and the full null arm was not executed (Section 7).

3.4 Protocol and suite

We score selection against the tool appropriate for the value in the model’s actual state and, for the primary gold-agreement analysis, score arguments against the canonical trajectory. Digit-only JSON strings are coerced to integers, so formatting is not a value error. Errors are syntactic (parse/schema failure) or semantic (executes but is wrong). Both arms use the same within-step retry budget: if a response fails to parse as a valid tool call (malformed JSON, missing required fields), the model is re-prompted with a structured error message up to two additional times; only the final attempt is scored as that step’s invocation, and earlier failed attempts are recorded as syntactic errors but do not count toward $\bar{p}_d$ or $\bar{g}_d$. If all retries fail, the step is scored as a syntactic error and the free-running arm continues with the last parseable state. This policy is identical in both arms. Conditional-on-state replay replaces the argument target with the argument implied by the model’s actual state, making the selection and argument criteria symmetric.

We evaluated five open-weight models available through one hosted API: llama-3.1-8b-instant, allam-2-7b, qwen3.6-27b, gpt-oss-120b, and llama-3.3-70b-versatile. Greedy decoding, fixed seeds, and caching by (model, calling mode, exact prompt) make each benchmark prompt deterministic. Provider constraints produced unequal nested sample sizes and prevented balanced family, proprietary-model, and calling-mode comparisons; we draw no claims along those axes.

Models run **nested prefixes** of one task suite, sized per model to its own measured token allowance. Task identity depends only on (depth, k , seed), never on how many tasks were requested, so cross-model contrasts on the shared prefix remain exactly paired while each model’s own estimate uses its full n .

4 Data Collected

All figures come from a dataset frozen at 14:09 on 2026-08-17; no missing model–depth cell is extrapolated.

Table 1: Per-model sample sizes at the freeze.

Model	tasks	d_1	d_2	d_4	d_6	d_8	status
llama-3.1-8b-instant	273	60	60	65	65	23	only model at depth 8
allam-2-7b	225	60	60	65	40	–	usable
gpt-oss-120b	108	26	26	28	28	–	usable
qwen3.6-27b	98	26	26	28	18	–	usable
llama-3.3-70b-versatile	49	12	12	13	12	–	usable, thin
gpt-oss-20b	0	–	–	–	–	–	excluded

gpt-oss-20b completed zero tasks after pilot calls consumed its daily allowance and is excluded from every empirical claim. Provider constraints explain the unequal sample sizes but are not treated as scientific findings.

5 Results

5.1 Propagation grows through depth 6 and exceeds baseline decay

Figure 2 and Table 2 show the core measurement. On llama-3.1-8b, $\bar{p}_d$ declines from 0.700 to 0.543 (22% relative), while $\bar{g}_d$ declines to 0.174 (75%). L_d rises sharply through depth 6 and is essentially unchanged at depth 8 (0.686 vs. 0.680); we therefore claim growth followed by a plateau, not strict monotonicity or significant adjacent-depth differences.

$L_1 = 0$ **exactly on every model**: both arms build an identical depth-1 prompt and share one cached response, providing an implementation invariant. At depth 6, llama-3.1-8b has identical baseline/free error counts at the first position (21/21), after which free-running errors rise while

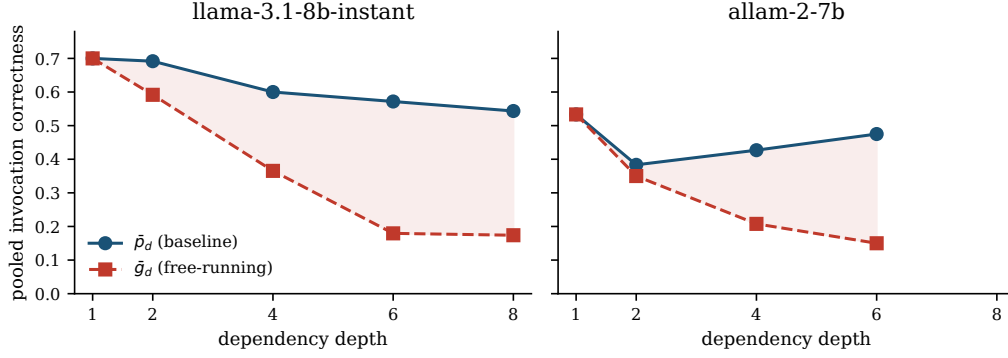


Figure 2: Pooled baseline correctness $\bar{p}_d$ versus pooled free-running correctness $\bar{g}_d$. The absolute gap widens through depth 6 and then plateaus for llama-3.1-8b; depth 1 is identical by construction.

Table 2: Invocation-pooled depth estimands for the two high- n models. Intervals are the reported 89% intervals from the frozen paired analysis.

llama-3.1-8b-instant					
depth d	$\bar{p}_d$	$\bar{g}_d$	L_d [89% interval]	calls	
1	0.700	0.700	0.000 [0.000, 0.000]	60	
2	0.692	0.592	0.145 [0.082, 0.222]	120	
4	0.600	0.365	0.391 [0.311, 0.474]	260	
6	0.572	0.179	0.686 [0.607, 0.766]	390	
8	0.543	0.174	0.680 [0.574, 0.772]	184	
allam-2-7b					
1	0.533	0.533	0.000 [0.000, 0.000]	60	
2	0.383	0.350	0.087 [0.022, 0.160]	120	
4	0.427	0.208	0.514 [0.431, 0.596]	260	
6	0.475	0.150	0.684 [0.624, 0.745]	240	

baseline errors remain approximately flat. The absolute gap and error-amplification summaries show the same through-depth-6 increase without dividing by baseline accuracy.

The lag association $\Delta_1 = P(k+1 \text{ correct} \mid k \text{ correct}) - P(k+1 \text{ correct} \mid k \text{ wrong})$ is positive for every model with errors (+0.32 to +1.00). It is descriptive rather than causal; its second term is zero under canonical gold agreement because of the unreachable-target mechanism below.

5.2 Ceiling models, and where in a chain a model errs

Three models sit at or near ceiling ($\bar{p}_d = 0.865\text{--}1.000$; Figure 3), so $L_d \approx 0$ is primarily a statement about task difficulty. On llama-3.3-70b-versatile at depth 4, 6 of 7 errors occur at a terminal position, leaving no downstream opportunity. At depth 6 its estimate is $L_6 = 0.190$ [0.062, 0.345], but the sample is thin. gpt-oss-120b has one fresh error, and qwen3.6-27b has zero errors across 298 free-running calls; neither supports a robustness claim.

An error at the terminal position cannot propagate, so a fixed-depth aggregate mixes error frequency with the remaining opportunity for downstream damage; $L_1 = 0$ is the limiting case. For llama-3.1-8b, terminal-error shares are 0.714 at $d = 2$ and 0.197 at $d = 6$, versus uniform-position baselines of 0.500 and 0.167. For llama-3.3-70b-versatile, the 6/7 terminal share at $d = 4$ is suggestive but extremely uncertain. Terminal-error share should therefore accompany L_d , and cross-model comparisons require comparable position distributions.

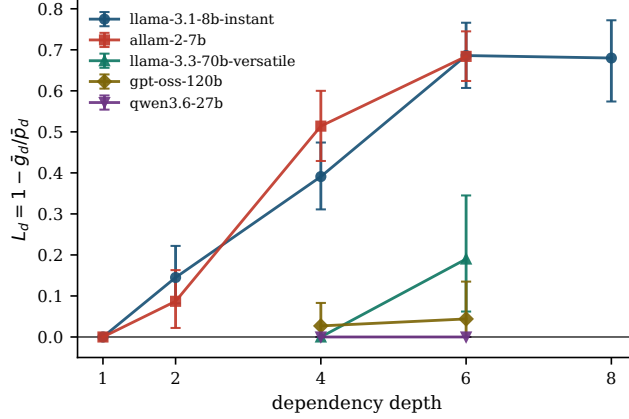


Figure 3: L_d versus depth for all usable models (89% intervals). Three models remain near ceiling; llama-3.3-70b-versatile shows a positive depth-6 estimate.

170 5.3 When canonical gold agreement ceases to measure behavior

171 Let H_k denote everything available to the agent before a post-divergence call, and let Y_k^* be the
172 canonical gold target used by the scorer. Define its conditional guessing probability

$$\epsilon_k(H_k) = \max_y P(Y_k^* = y \mid H_k). \quad (4)$$

173 **Proposition (unreachable-target bound).** For any possibly randomized agent whose prediction
174 $\hat{Y}_k$ depends only on H_k ,

$$P(\hat{Y}_k = Y_k^* \mid H_k) \leq \epsilon_k(H_k). \quad (5)$$

175 Consequently, if q_k^{gold} is correctness in corrupted context and $p_k > 0$ is clean-context correctness,
176 gold-defined severity satisfies $\pi_k^{\text{gold}} = 1 - q_k^{\text{gold}}/p_k \geq 1 - \mathbb{E}[\epsilon_k]/p_k$. Across recovery opportunities
177 $\mathcal{K}$, the probability of any chance reconvergence is at most $\sum_{k \in \mathcal{K}} \mathbb{E}[\epsilon_k]$.

178 *Proof.* Conditional on H_k , sum over candidate outputs: $P(\hat{Y}_k = Y_k^* \mid H_k) = \sum_y P(\hat{Y}_k = y \mid$
179 $H_k)P(Y_k^* = y \mid H_k) \leq \max_y P(Y_k^* = y \mid H_k)$. The recovery statement is the union bound. $\square$

180 The proposition is about *construct validity*: when ϵ_k is negligible, gold agreement cannot distinguish
181 a locally competent continuation from arbitrary behavior. It does not say the operational statistic is
182 numerically unidentified; rather, the statistic is pinned near a boundary that no longer represents the
183 intended behavioral severity.

184 **Instantiation in this suite.** After divergence, canonical arguments depend on the hidden-constant
185 output associated with the gold state. Under the generator assumption that this target is conditionally
186 uniform over $M = 100,000$ values, $\epsilon_k = 10^{-5}$. We observe 0 gold matches among 869 corrupted-
187 context calls and 0 canonical reconvergences among 580 opportunities. Even a pure-guess process has
188 expected reconvergences $580/M = 0.0058$, with probability of any such event at most 0.0058. Zero
189 therefore carries essentially no information about whether the agent can apply the local routing/copy
190 rule in its actual state.

191 If one nevertheless substitutes the scorer-implied boundary $\pi = 1$ and zero canonical recovery into the
192 step model, it collapses to $g_k^{\text{model}} = c_k p_k$ and $c_k = \prod_{j < k} p_j$. A hierarchical fit to pooled aggregates
193 returned severity posteriors of 0.92 and 0.73 despite zero observed post-divergence gold matches;
194 $\hat{R} \leq 1.0005$, $\text{ESS} \geq 5,823$, and zero divergences show only that the misspecified posterior was
195 explored successfully: a targeted posterior predictive check finds 0 of 6,000 replicates reproducing
196 the observed zero canonical matches among corrupted-context calls, against a predicted mean of 67.7.

197 **Counterexample and scope.** Fixed reference is not sufficient for the proposition. If Y_k^* is re-
198 constructible from the observable history, independent of the corrupted value, or reachable through

reset/re-query actions, ϵ_k may be large and canonical recovery can be meaningful. Final-state unit tests and milestone-DAG evaluators can also accept alternative valid trajectories. Our claim is therefore restricted to canonical scoring under an empirically or structurally small ϵ_k ; it is not a blanket claim about execution match, AST match, or fixed references.

5.4 A remedy: conditional-on-state scoring

Credit a call when it *correctly continues from the value the model actually holds*, not when it matches the canonical state. Routing selection was already computed conditionally; replay adds the argument-side counterpart. Because every raw completion was cached, this required 881 cache reads and no new model queries.

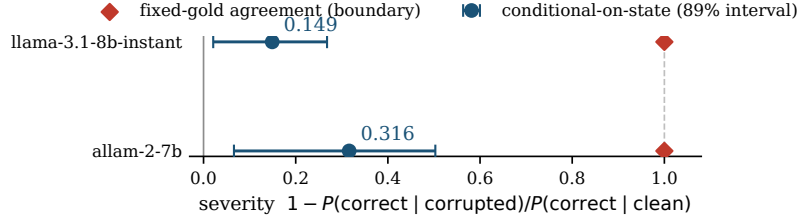


Figure 4: Gold-agreement severity is pinned at its boundary, whereas conditional-on-state replay yields interior estimates (reported 89% intervals). The contrast is a change of estimand, not an improved fit to the same target.

Clean-context calls score identically under both rules (0.658 vs. 0.658), an implementation invariant. Conditional severity is 0.149 [0.021, 0.268] for llama-3.1-8b and 0.316 [0.066, 0.503] for allam-2-7b. Thus state corruption is associated with genuine loss of local rule application, but far less than the boundary value assigned by canonical gold agreement. Evidence is limited to two models and a weak copy-argument competence test. A controlled error-injection check on allam-2-7b alone finds a smaller causal estimate, $\hat{\pi} = +0.108 [+0.006, +0.215]$ for selection-channel corruption (argument-channel corruption showed no effect), indicating the observational estimate above is inflated by roughly $2.9\times$: the free-running arm lets the model choose which state to corrupt, while the injection arm assigns it, and only the latter isolates a causal effect.

Trade-off. Conditional scoring requires an executable transition model for the agent’s actual state and does not imply end-task success. Canonical gold, conditional-on-state, milestone, and final-state scores answer different questions; none should be presented as a universal replacement.

5.5 Rule-following is measurable directly, and aggregates hide it

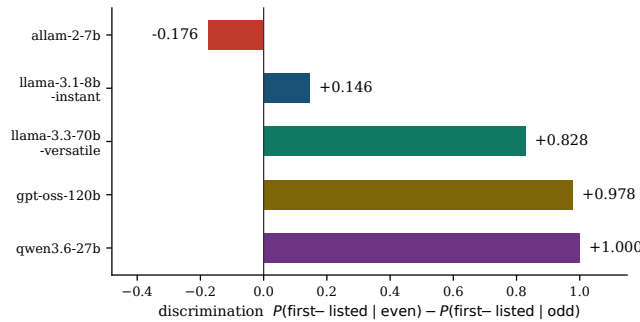


Figure 5: Discrimination, defined as $P(\text{first-listed} | \text{even}) - P(\text{first-listed} | \text{odd})$, per model. Order matches the reliability ranking of Section 5.1 (both computed from the same routing data).

Define **discrimination** as $P(\text{first-listed} | \text{ref even}) - P(\text{first-listed} | \text{ref odd})$: 0 means the rule is ignored, +1 perfect application, negative means *anti*-correlated. Overall first-listed rates (0.478–0.497, four of five models) are indistinguishable and consistent with no position bias whatever; the

224 *same data*, conditioned on parity, resolves discrimination across the full range (Figure 5), and the
 225 order matches reliability established independently in Section 5.1. **Discrimination is a per-model**
 226 **diagnostic of whether a model performs the task at all, invisible to any metric aggregated over**
 227 **the conditioning variable.**

228 Three qualitatively different relationships, not one spectrum: `qwen3.6-27b` *perfectly follows* the rule
 229 (+1.000, zero errors in 298 steps); `llama-3.1-8b` *weakly follows* it (+0.146, $z=4.7$); `allam-2-7b`
 230 is *anti-correlated* (-0.176 , $z=-5.0$): it picks the first-listed tool *more* on odd refs (0.774) than even
 231 (0.597), where its accuracy is worst (0.223 vs. 0.595). `allam-2-7b`’s propagation loss is therefore
 232 not “a weak model that errs more” but a model *not performing the routing task*.

233 5.6 Two negative results

234 **Untestable, not underpowered:** a hypothesised error-composition shift along the chain cannot be
 235 tested here because **389 of 389** clean-context errors are selection errors: on a copy-argument task
 236 there are only two ways to fail cleanly, mis-apply the parity rule (hard) or mis-transcribe a verbatim
 237 number (models never do). **Unresolved by design:** the suite offers two 2-point scale contrasts rather
 238 than 3–4, not comparable in kind (`llama 8B`→`70B` dense, $8.75\times$; `gpt-oss 20B`→`120B`, $6\times$ total
 239 but $\approx 1.4\times$ active params). We report both axes and draw no scale conclusion, since the larger
 240 models sit at ceiling on these tasks.

241 6 Pipeline Validation, and What Simulation Cannot Establish

242 Before spending API budget we validated the full pipeline against six simulated policies with known
 243 ground truth, eliminating seven harness artifacts (retry-budget asymmetry, selection errors mis-
 244 attributed under poisoning, a contaminated syntactic-share estimator, among others). All seven were
 245 discovered and fixed before any real-model data was collected; 14 regression-test modules lock the
 246 corrected behavior and pass on the frozen dataset. Two further artifacts were not coding errors, which
 247 is what makes them instructive: comparing carried value against *expected argument* rather than
 248 *previous gold output* is correct on a copy-argument task and silently wrong under any transform; and
 249 the simulated policies recovered by reading ground truth directly, out of band, through a channel no
 250 real model has. Correct code, correct configuration, invalid measurement. The second artifact is why
 251 the recovery channel was never estimable from observable data in any version of this pipeline. The
 252 81% parser-failure finding for `qwen3.6-27b` was discovered during post-collection cache replay; the
 253 parser was fixed, cached completions were re-scored (not re-collected), and no completion changed
 254 its parsed value (the failure mode was clean breakage, not silent corruption).

255 Three principles follow, nested by scope: **(1) simulated validation** certifies only the channels it
 256 exercises and silently passes the ones it does not (our mock backend read state from stashed context
 257 rather than the message history, so an agent loop that never threaded observations into the conversation
 258 passed the entire simulated suite); **(2) a parser or scorer can silently flip the direction of a result**,
 259 not just its precision: replaying cached completions through our pre-fix extractor, **319/394 (81.0%)** of
 260 `qwen3.6-27b`’s responses would have failed to parse; **(3) every constant estimated from one model**
 261 **or assumed for convenience should be verified per-model:** a shared output-token constant hid
 262 `qwen`’s true cost ($6.6\times$ higher); an assumption that parity was incidental hid `allam`’s anti-correlation;
 263 per-depth pooling hid where each model’s errors cluster.

264 This establishes that the estimation and aggregation pipeline recovers configured parameters. It does
 265 **not** establish the prompt-construction and observation-passing path, and no out-of-band simulation
 266 can.

267 7 Limitations and Scope

268 This study uses synthetic integer-argument tasks, giving exact state transitions at the cost of ecological
 269 validity. The strongest empirical findings are carried by two 7–8B models; three stronger models
 270 are largely at ceiling. The unreachable-target proposition is general only through its explicit ϵ_k
 271 condition, while the empirical claim that our generator makes ϵ_k negligible depends on its hidden-
 272 state construction. Validation on natural-language arguments and stateful environments such as
 273 ToolSandbox or AppWorld is required before claiming prevalence in deployed agents.

Three designed conditions were not executed: transformed arguments, randomized presentation order, and a calling-mode ablation. A linear-task control was executed on allam-2-7b alone: at depth 6, $L_6 = 0.156$ [0.089, 0.229] against 0.684 [0.624, 0.745] on routing. The linear variant removes selection dependency but retains value copying; the residual loss is substantially smaller than routing’s but excludes zero at depths 4 and 6, indicating that the baseline/free-running contrast captures effects beyond selection-error propagation alone. These are material threats, not cosmetic extensions. In particular, allam-2-7b’s anti-correlation may depend on the parity/order presentation, and copy arguments do not test semantic argument construction. Two of the five original models (llama-3.1-8b-instant, llama-3.3-70b-versatile) were retired from the hosting provider within ten days of data collection, preventing any new ablation conditions from being compared against their primary-arm baselines; this is the third instance of undocumented infrastructure instability encountered in this study. The unequal nested sample sizes preclude scale or family comparisons. The reported intervals require the task-level artifact for reproduction; call-level binomial intervals would be invalid because steps within a task are dependent.

8 Conclusion

Matched baseline/free-running evaluation reveals large propagation on two small models: at depth 6, pooled free-running correctness is about 68% below each model’s baseline rate. Absolute excess error and error amplification show the same pattern, while stronger models are mostly uninformative because the tasks are too easy.

The broader result is conditional. When the canonical next target has negligible predictability from the post-divergence observable history, any gold scorer credits local behavior only at that guessing probability. Gold-defined severity then approaches a scorer-imposed boundary, and canonical reconvergence does not measure behavioral recovery. Conditional-on-state replay changes the estimand to local continuation quality and produces interior estimates in this suite.

The practical rule is narrower and testable: **before interpreting post-divergence gold agreement, estimate whether the gold target remains reconstructible from what the agent can observe.** If not, report state-conditional competence separately from milestone or final-state success.

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A Aggregate-data sensitivity checks

Table 3 reports the aggregate correct/total counts underlying Table 2, together with absolute excess error A_d and error amplification E_d . Task-cluster confidence intervals require the per-task records in the release artifact; the deterministic summaries below do not.

Table 3: Aggregate counts and alternative fit-free propagation summaries. $A_d = \bar{p}_d - \bar{g}_d$; $E_d = (1 - \bar{g}_d)/(1 - \bar{p}_d)$.

Model	d	baseline	free-running	A_d	E_d
llama-3.1-8b	1	42/60	42/60	0.000	1.000
	2	83/120	71/120	0.100	1.324
	4	156/260	95/260	0.235	1.587
	6	223/390	70/390	0.392	1.916
	8	100/184	32/184	0.370	1.810
allam-2-7b	1	32/60	32/60	0.000	1.000
	2	46/120	42/120	0.033	1.054
	4	111/260	54/260	0.219	1.383
	6	114/240	36/240	0.325	1.619

B What each scoring regime identifies

Table 4: Scoring regimes answer complementary questions.

Scorer	Alternative paths	Primary estimand	Post-divergence limitation
Canonical call/trajectory match	Usually no	Agreement with one reference	Local competence is confounded when the target has small ϵ_k
Conditional-on-state (ours)	Yes, one state at a time	Correct local continuation	Does not imply final task success
Milestone DAG (ToolSandbox)	Yes	Intermediate/final progress	Attribution depends on milestone coverage
Final-state tests (AppWorld, τ -bench)	Yes	End-task state and side effects	Usually lacks per-call diagnosis

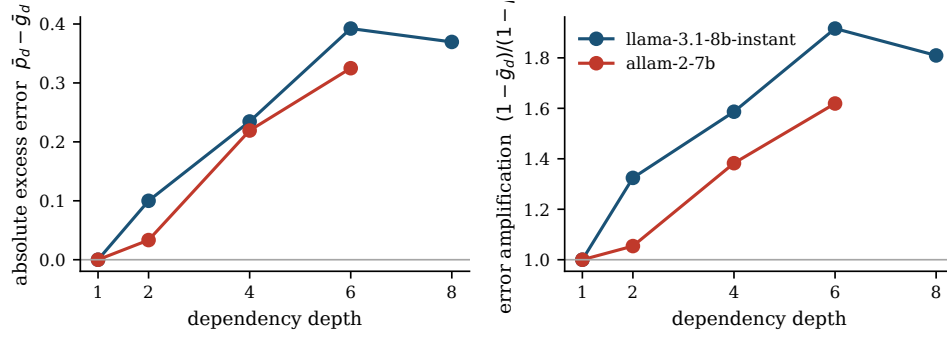


Figure 6: The propagation pattern is not specific to $L_d = 1 - \bar{g}_d/\bar{p}_d$. Absolute excess error grows to 0.392 and 0.325 by depth 6; free-running errors are amplified by factors of 1.916 and 1.619 relative to baseline errors. Both metrics plateau or terminate after depth 6 rather than supporting strict monotonicity.

347 C Posterior predictive check for the step-model fit

348 Section 5.3 states that the hierarchical fit is misspecified rather than merely uncertain. This appendix
 349 shows the check behind that claim, and the reason a conventional diagnostic does not detect it.

350 The fitted model implies that a call issued on a corrupted context succeeds with probability
 351 $(1 - \pi)p_k$, which is strictly positive for any $\pi < 1$. Summed over the corrupted-context calls
 352 in the suite, it therefore predicts a positive number of canonical matches. The observed number
 353 is zero. We rebuilt the fitted model with one additional observed node for that subset and drew
 354 from `pm.sample_posterior_predictive` against the existing trace; no new model queries were
 355 required.

Table 5: Posterior predictive checks on two fits. The targeted check evaluates canonical matches among corrupted-context calls only; the aggregate check evaluates total free-running successes, which is the quantity such a fit is ordinarily judged on. Both fits pass the aggregate check and fail the targeted one.

	Real-model fit	Simulated-suite fit (control)
Corrupted-context calls	869	9,624
Observed canonical matches	0	1,648
Predicted mean	67.7	3,173
Predicted 89% interval	[39, 98]	[2,899, 3,427]
Minimum over 6,000 replicates	10	2,427
Targeted check, $\Pr(\text{rep} \leq \text{obs})$	0.000	0.000
Aggregate check, Bayesian p	0.420	0.282

356 Two points follow. First, **the conventional check would have certified this fit as adequate.**
 357 Replicated total free-running successes match the observation closely ($p = 0.420$), because the
 358 marginal rate g_k averages over clean and corrupted contexts and the clean ones dominate the counts.
 359 A posterior predictive check aimed at the aggregate is structurally blind to a failure confined to the
 360 corrupted subset, so passing it is not evidence that the step model describes post-divergence behavior.
 361 Reporting an aggregate predictive check alongside boundary-valued severity posteriors is therefore
 362 not a safeguard.

363 Second, the same targeted check fails on the simulated suite, where the generating policies *were*
 364 configured to recover and the observed count is a substantial 1,648. The fit over-predicts there too,
 365 by roughly a factor of two. The over-prediction is thus a property of the fit, which under-estimates
 366 severity so $(1 - \pi)$ is too large, and not an artifact peculiar to real data sitting at a scorer-imposed
 367 boundary. This is what distinguishes the finding from a claim that real models are simply harder to fit.

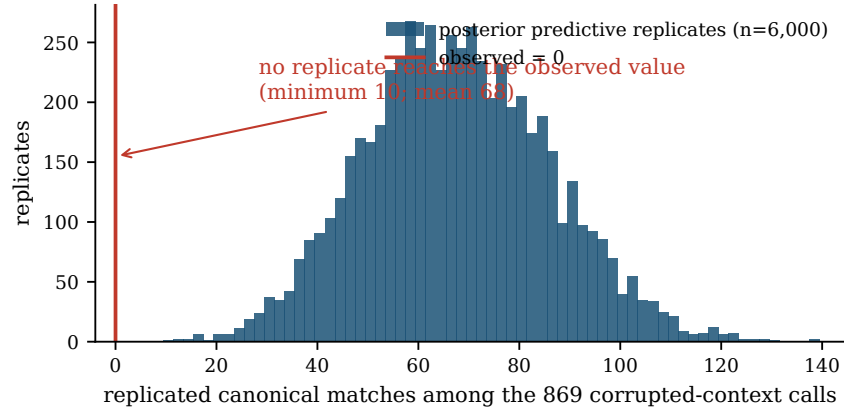


Figure 7: Posterior predictive distribution of canonical matches among the 869 corrupted-context calls in the real-model fit, against the observed value. No replicate reaches the observation; the smallest of 6,000 is 10.

368 D Linear-task control arm

369 The linear variant announces the tool order in the prompt, so the correct tool at each step is fixed in
 370 advance and cannot depend on the value carried in. What it retains is argument-value copying: each
 371 argument is a verbatim copy of the previous observation, so a wrong value still propagates forward.
 372 The arm was executed on `allam-2-7b` at the routing arm’s depths and sample sizes. Depth 6 stopped
 373 on an exhausted daily token allowance at 31 of 40 planned tasks; the achieved counts are a nested
 374 prefix and are used as such.

Table 6: Invocation-pooled depth estimands for `allam-2-7b` on both task families. Intervals are 89% paired bootstrap intervals over tasks, the same estimator and convention used for the routing arm.

Linear (selection dependency removed)				
depth d	$\bar{p}_d$	$\bar{g}_d$	L_d [89% interval]	calls
1	0.950	0.950	0.000 [0.000, 0.000]	60
2	0.992	0.983	0.008 [0.000, 0.026]	120
4	0.962	0.908	0.056 [0.028, 0.086]	260
6	0.930	0.785	0.156 [0.089, 0.229]	186
Routing (selection dependency present)				
1	0.533	0.533	0.000 [0.000, 0.000]	60
2	0.383	0.350	0.087 [0.022, 0.160]	120
4	0.427	0.208	0.514 [0.431, 0.596]	260
6	0.475	0.150	0.684 [0.624, 0.745]	240

375 Linear tasks remove the selection dependency but retain argument-value copying, so the two arms
 376 differ in whether an upstream error can change which tool is correct downstream, not in whether
 377 a wrong value is carried forward. At depth 6 the routing loss is $4.4\times$ the linear one (0.684 against
 378 0.156), which is consistent with selection-error propagation accounting for the majority of total loss.
 379 The linear intervals at depths 4 and 6 both exclude zero, consistent with the baseline/free-running
 380 contrast capturing effects beyond selection-error propagation, though the specific mechanism is not
 381 resolved by this control alone; the linear arm is therefore a bound on the selection dependency’s
 382 contribution rather than a null. The two families share the harness, the scorer and the prompt length
 383 at each depth, so neither of those explains the gap between them.

E Error-injection ablation

To test whether the observational propagation estimate reflects a causal effect of state corruption rather than an arm-construction confound, we injected controlled errors into otherwise identical trajectories for allam-2-7b (52 of 60 planned tasks, 316 pairs). At a specified position $j \in \{1, 3, 5\}$, two branches diverge from the same prefix: a *corrupted* branch where one tool output is replaced with a wrong value, and a *corrected* branch where the original value is preserved. Both branches then continue through the same model. Because both share an identical prefix and differ in exactly one message, the contrast isolates state corruption from prompt-style, retry-history, and demonstration-quality differences.

Validity check. The corrected branch reproduces the baseline arm’s accuracy exactly across all 40 overlapping tasks at every position tested (+0.000 at $j = 1, 3, 5$), confirming the injection mechanism introduces no construction artifact.

Two injection modes. `parity_flip` changes the carried value’s parity, altering which tool the routing rule selects (a selection-channel corruption). `parity_preserving` changes the value without changing its parity, so the correct tool is unchanged but the argument is wrong (an argument-channel corruption).

Table 7: Causal severity from controlled error injection on allam-2-7b, scored conditional-on-state. Intervals are 89% paired bootstrap intervals resampling pairs. Discordant counts are the pairs in which exactly one branch was correct: b favours the corrected branch, c the corrupted one.

condition	corrected	corrupted	$\hat{\pi}$ [89% interval]	b/c
<code>parity_flip</code> (selection)	0.411	0.304	+0.108 [+0.006, +0.215]	65/48
<code>parity_preserving</code> (argument)	0.411	0.443	−0.032 [−0.057, −0.013]	0/5
observational (routing free arm)	n/a	n/a	+0.316 [+0.066, +0.503]	n/a

Three findings follow. First, the observational estimate is biased upward by approximately $2.9\times$ (+0.316 against +0.108), and the observational value lies outside the causal interval. In the free-running arm the model selects its own corruptions, so whatever produced the upstream error remains present downstream; the injected contrast removes that dependence by assigning corruption externally. Second, the effect is confined to the selection channel: `parity_preserving` corruption produced no degradation at any position, with zero pairs favouring the corrected branch, so a wrong argument value alone does not measurably impair this model’s next call. Third, severity is not constant along the chain. The position profile is +0.132 at $j = 1$, +0.226 at $j = 3$ and −0.038 at $j = 5$, of which only $j = 3$ excludes zero. The step model of Section 3.2 treats π as position-independent, and these estimates do not support that assumption.

These results are from a single model at one depth and a truncated allocation. The pooled selection-channel interval only narrowly excludes zero and one of three positions carries the effect, so the ablation establishes the direction of the observational bias more firmly than it establishes the magnitude of the causal severity. The eight tasks not collected would not change that. We report it as a single-model causal check on the estimand, not as suite-wide validation.